

1.解: $L = \{ (1, 2), (1, 3), (1, 6), (2, 3), (2, 6), (3, 6), (1, 1), (2, 2), (3, 3), (6, 6) \}$

$D = \{ (1, 2), (1, 3), (1, 6), (2, 6), (3, 6), (1, 1), (2, 2), (3, 3), (6, 6) \}$

则 $L \cap D = \{ (1, 2), (1, 3), (1, 6), (2, 6), (3, 6), (1, 1), (2, 2), (3, 3), (6, 6) \}$, 即 $|L \cap D| = 9$.

2.空集; 全集; 全集; 空集

3.

$$(n^{n(n-1)} - n^{\frac{n(n-1)}{2}})(n^n - 1)$$